

EIOT Project Document

Project Title: Smart Public Toilet Monitoring and Maintenance System Using IoT

S.no	Team Members
1.	Santhosh P
2.	Kishore Kumar G
3.	Mohan G
4.	Manikandaprabhu G A
5.	Prasanth Dharsan J

Abstract:

Public toilets in high-traffic areas often suffer from **poor hygiene, water wastage, and delayed maintenance due to the lack of real-time monitoring**. This project presents a **Smart Public Toilet Monitoring and Maintenance System** that uses embedded systems and IoT technologies to ensure cleanliness, efficient water usage, and timely maintenance. The system automatically flushes water before and after toilet usage, monitors environmental conditions to estimate Odor, tracks usage count, and alerts maintenance personnel when cleaning is required. Visual indicators, audio alerts, and a user complaint mechanism improve usability and responsiveness. This solution enhances public sanitation and supports smart city initiatives.

Objective:

The main objectives of this project are:

- To ensure toilet cleanliness by automatic flushing before and after use
- To monitor toilet usage and environmental conditions in real time
- To indicate toilet status using LED indicators
- To provide immediate alerts for maintenance requirements
- To allow users to manually raise complaints when necessary
- To display toilet status information to users through an LCD
- To collect usage data for maintenance planning and analysis

Process Overview:

1. When a user enters the toilet, the **PIR sensor** and **ultrasonic sensor** detect human presence.
2. The system performs an **automatic pre-flush** to ensure cleanliness.
3. During usage, the toilet status is indicated as **Occupied** using a yellow LED and LCD message.
4. After the user exits, the system triggers an **automatic post-flush** through a relay-controlled water motor.
5. The **DHT11 sensor** monitors humidity to estimate odor conditions.
6. If poor hygiene conditions are detected, a **red LED and buzzer** alert maintenance staff.

7. A **complaint button** allows users to manually report issues.
8. Usage count and status data are sent to an Telegram Bot for monitoring and analysis.

Applications:

- Public toilets in railway stations and bus terminals
- Airports and shopping malls
- Educational institutions
- Hospitals and public buildings
- Smart city sanitation systems

Components Required

S.No	Component Name
1	ESP32 Development Board
2	PIR Motion Sensor
3	Ultrasonic Sensor (HC-SR04)
4	DHT11 Temperature & Humidity Sensor
5	Relay Module (5V)
6	Water Pump / Motor
7	16×2 LCD Display
8	LED Indicators (Red, Yellow, Green)
9	Buzzer
10	Push Button (Complaint Button)
11	Connecting Wires
12	Breadboard / PCB
13	Power Supply

S.No	Component Name
14	MQ-2 gas sensor
15	Air Humidifier

Estimated Cost (Reference: Robu.in)

Component	Approx. Price (₹)
ESP32 Development Board	450
PIR Sensor	120
Ultrasonic Sensor	110
DHT11 Sensor	90
Relay Module	80
MQ-4 sensor	80
Water Pump / Motor	250
Humidifier	160
16×2 LCD Display	180
LEDs (Set)	30
Buzzer	40
Push Button	20
Wires & Breadboard	150
Total Estimated Cost	≈ 1,760